The Wave Equation

Exam code: 0625 & 0972

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The wave equation

- The equation used to calculate wave speed is:

$$v = f \times \lambda$$

- Where:
  - v = wave speed, measured in metres per second (m/s)
  - f = wave frequency, measured in hertz (Hz)
  - λ = wavelength, measured in metres (m)
- Wave speed is defined as:

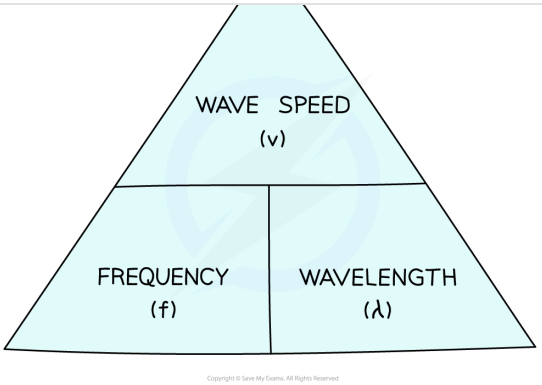
The distance travelled by a wave each second

- Wave speed is the speed at which energy is transferred through a medium
- Transverse and longitudinal waves both obey the wave equation

Wave speed formula triangle

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- For more information on how to use a formula triangle, refer to the revision note on speed & velocity

Frequency and period

- The period of a wave is defined as:

The time taken for one complete oscillation to pass a fixed point

- The frequency of a wave is related to its period by the following equation:

$$f = \frac{1}{T}$$

- And therefore,

$$T = \frac{1}{f}$$

- Where:
  - f = frequency, measured in Hz
  - T = period, measured in s

Worked Example

A wave in a pond has a speed of 0.15 m/s and a time period of 2 seconds. Calculate:

a) The frequency of the wave

b) The wavelength of the wave

Answer:

Part (a)

Step 1: List the known quantities

- Time period, T = 2 s

Step 2: State the equation relating time period and frequency

Step 3: Rearrange for frequency, f, and calculate the answer

$$f = \frac{1}{T} = \frac{1}{2}$$

Frequency, f = **0.5 Hz**

Part (b)

Step 1: List the known quantities

- Wave speed, v = 0.15 m/s
- Frequency, f = 0.5 Hz

Step 2: Write out the wave speed equation

$$v = f \times \lambda$$

Step 3: Rearrange the equation to calculate the wavelength

$$\lambda = \frac{v}{f}$$

Step 4: Use the frequency you calculated in part (a) and put the values into the equation

$$\lambda = \frac{0.15}{0.5}$$

Wavelength, λ = **0.30 m**

Examiner Tips and Tricks

When stating equations make sure you use the right letters. For example, use λ for wavelength, not L or W

If you can't remember the correct letters, then just state the word equations

Be careful with units: wavelength is usually measured in metres and speed in m/s, but if the wavelength is given in cm you might have to provide the speed in cm/s

Likewise, watch out for the frequency given in kHz: 1 kHz = 1000 Hz

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